

- **SQL Traffic Trends (Task 1):** Compares total yearly traffic volumes across 2012–2017, documenting notable system decreases (e.g., the 2014–2015 construction drops) and evaluating holiday temperature changes (New Year's vs. Labor Day) to trace ambient shifts in seasonal regional workflows.
- **Descriptive Statistics & Correlation (Task 2):** Highlights the calculated mean, median, standard deviation, and variance profile of the 48,000 traffic records. It documents the core correlation coefficient between ambient temperature and vehicle counts, providing a brief explanation of why this physical relationship does not imply direct behavioral causation.
- **Probability & Congestion Splits (Task 3):** Details the exact baseline probability of hitting severe congestion (traffic volume > 5,500 vehicles) and explores the conditional dependencies between clear weather and traffic gridlock to statistically evaluate corridor operational independence.
- **Power BI Visual Analytics (Task 4):** Summarises the key insights derived from the data preparation pipeline (converting Kelvin inputs to Celsius, extracting hour columns, and creating Low/Medium/High volume groups), detailing peak transit windows (07:00 and 16:00) and specific weather categories that record the highest average vehicle flows.